

Abstract

This project presents a comprehensive implementation of a **Football Player Performance Prediction System**, leveraging advanced machine learning and interactive tools to deliver actionable insights into player attributes and performance. The system integrates a Random Forest Regressor model trained on real-world football data to predict a player's overall rating based on key attributes, such as physical, mental, and technical scores.

The project incorporates a **Flask**-based backend for model hosting and prediction services and a responsive, user-friendly **Streamlit** front-end interface for interaction.

Key features include:

1. **Feature Engineering:** Attributes like Physical, Passing, Mental, and Shooting Scores are dynamically computed based on individual metrics, enhancing the model's interpretability and prediction accuracy.
2. **Machine Learning Model:** The Random Forest Regressor, optimized through cross-validation and hyperparameter tuning, ensures robust and accurate predictions, achieving high R^2 , low Mean Absolute Error (MAE), and low Mean Squared Error (MSE).
3. **Dynamic Chatbot Integration:** A custom AI chatbot provides contextual explanations, helping users understand player predictions and explore hypothetical scenarios.
4. **CI/CD Pipeline (Partial):** Automated deployment pipelines enable seamless integration, testing, and deployment of updates on platforms like **Docker** and **Google Cloud Platform** (GCP).
5. **Interactive Interface:** The **Streamlit** application allows users to input player attributes and visualize predictions.

This project showcases the potential of AI-driven decision-making tools in enhancing player scouting, team selection, and performance analysis in the football industry. The application is designed to be scalable, modular, and accessible, providing a template for similar predictive analytics projects in other domains.